

09_API_Endpoint_Spec.md

MonitorAI v1.5

FileMaker Server 19 Data API Specification

Version: 1.5 REV2

Status: FINAL BASELINE

1. Purpose

This document defines all communication between:

MonitorAI iPhone App

and

FileMaker Server 19

using the FileMaker Data API.

The API layer must preserve:

- Image Integrity
 - Metadata Integrity
 - Session Traceability
 - AI Dataset Quality
-

2. Environment

Backend

FileMaker Server 19

Database

MonitorAI

API

FileMaker Data API

Transport

HTTPS Only

Authentication

Bearer Token

Format

JSON

3. Base URL

Example

`https://monitor.example.com`

Data API Root

`/fmi/data/v1`

Note:

Use the version enabled on FileMaker Server 19.

4. Authentication

Login

POST

/fmi/data/v1/databases/MonitorAI/sessions

Response

token

Store

iOS Keychain

Logout

DELETE

/fmi/data/v1/databases/MonitorAI/sessions/{token}

5. API Layouts

Required Layouts

MonitorAI_API_Sessions

MonitorAI_API_Captures

MonitorAI_API_Metadata

MonitorAI_API_UploadQueue

MonitorAI_API_TrainingLabels

MonitorAI_API_SystemSettings

6. Session Creation

Layout

MonitorAI_API_Sessions

Endpoint

POST

/fmi/data/v1/databases/MonitorAI/layouts/MonitorAI_API_Sessions/records

Required Fields

SessionUUID

PatientUUID

SessionStartTime

DeviceModel

DeviceModelIdentifier

iOSVersion

AppVersion

Response

recordId

modId

Store Both

recordId

modId

for future updates.

7. Capture Creation

Layout

MonitorAI_API_Captures

Endpoint

POST

/fmi/data/v1/databases/MonitorAI/layouts/MonitorAI_API_Captures/records

Required Fields

CaptureUUID

SessionUUID

CaptureStep

CaptureName

CaptureStartTime

CaptureSuccessTime

CaptureDurationSeconds

RetakeVersion

FinalAccepted

Response

recordId

modId

Store

CaptureRecordID

CaptureModID

8. Metadata Creation

IMPORTANT

Metadata must be created BEFORE image upload.

Recommended Order

Session

↓

Capture

↓

Metadata

↓

Image Upload

Reason

If image upload fails, metadata already exists.

No capture context is lost.

Endpoint

POST

/fmi/data/v1/databases/MonitorAI/layouts/MonitorAI_API_Metadata/records

Required Fields

MetadataUUID

CaptureUUID

PTG

PTGCurrentDot

PTGMaxDot

Visibility

VisibilityScore

MaxVisibilityScore

BuccalVisibilityScore

OcclusalVisibilityScore

Capture

CaptureTriggerType

PlateauTriggered

PlateauDurationSeconds

Image Quality

FocusScore

BlurScore

ExposureScore

CompositionScore

Calibration

CalibrationOffsetX

CalibrationOffsetY

CalibrationQualityScore

Lighting

TorchEnabled

TorchLevel

ExposureValue

BrightnessScore

WhiteBalanceEstimate

Timing

UploadStartTime

UploadCompleteTime

UploadDurationSeconds

Response

recordId

modId

9. Full Resolution Image Upload

Layout

MonitorAI_API_Captures

Container Field

ImageFile

Endpoint

POST

/fmi/data/v1/databases/MonitorAI/layouts/MonitorAI_API_Captures/records/{recordId}/containers/
ImageFile/1

Upload Type

multipart/form-data

10. Thumbnail Upload

Container Field

ThumbnailFile

Endpoint

POST

/fmi/data/v1/databases/MonitorAI/layouts/MonitorAI_API_Captures/records/{recordId}/containers/
ThumbnailFile/1

11. Image Dataset Metadata

Recommended Additional Fields

Store inside CaptureImages

ImageWidth

ImageHeight

ImageSizeBytes

ImageSHA256

Purpose

Dataset validation

Duplicate detection

Integrity verification

Future AI export

12. Upload Queue

Layout

MonitorAI_API_UploadQueue

Endpoint

POST

/fmi/data/v1/databases/MonitorAI/layouts/MonitorAI_API_UploadQueue/records

Required Fields

QueueUUID

CaptureUUID

RetryCount

QueueStatus

LastRetryTime

LastErrorCode

LastErrorMessage

13. Session Completion

Layout

MonitorAI_API_Sessions

Endpoint

PATCH

/fmi/data/v1/databases/MonitorAI/layouts/MonitorAI_API_Sessions/records/{recordId}

Fields

SessionEndTime

SessionDurationSeconds

UploadStatus

UploadStatus

Uploaded

14. Retry Strategy

Automatic Retry

1 minute

↓

5 minutes

↓

15 minutes

↓

1 hour



Manual Retry

Store

RetryCount

LastRetryTime

LastErrorCode

LastErrorMessage

15. Error Handling

401

Authentication Failure



Re-login

Network Failure



Offline Queue

Container Upload Failure



Retry Queue

5xx

Server Failure



Retry Later

16. Security Requirements

Required

HTTPS

TLS 1.2+

Bearer Token

Keychain Storage

Prohibited

Guest Access

Shared Admin Accounts

Plain Text Credentials

Patient Data In Logs

17. Dataset Integrity Requirements

Every accepted image must have:

1 Capture Record

+

1 Metadata Record

+

1 Full Resolution Image

+

1 Thumbnail

Every uploaded image must be traceable to:

Patient

↓

Session

↓

Capture

↓

Metadata

18. Final Principle

FileMaker Data API is not only a transport layer.

It is the mechanism that preserves:

- Dataset Integrity
- Clinical Traceability
- Future AI Value

No upload process should compromise these goals.